

mol

Mixture of Limits — Navigation Law for Computer Intelligence

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Measurement. Research study, not a journal final. Energy figures from the software reference are OpCounter analytical estimates, not board power. We have not synthesized or metered an FPGA board. No fabricated citations.

Floors where more bits stop buying outcomes. Compression and predictive formulas beat excess enumeration; neural nets are a demoted residual leaf. Soft-ref constructive existence in software — not FPGA or board package energy.

Abstract

Mixture of Limits (MoL) is a navigation law for computer intelligence: there exist **floors** past which additional tokens, parameters, or joules do not purchase verifiable progress on a task coordinate. The law is information-theoretic and rooted in physics. Across human scientific history, compression into predictive formulas and invariants has repeatedly beaten excess enumeration of observations — from Kepler’s laws over Tycho’s tables, through Newton’s closed forms, to Shannon’s bit accounting and Landauer’s thermodynamic floor on irreversible erasure. MoL operationalizes that lineage for machines: **Lookup** → **Formula** → **Solver/settle** → **Model LAST**, with close owned as propose → certify → commit/refuse → receipt.

The industry default escalates Mixture-of-Experts (MoE) capacity *inside* a generative corridor. MoL instead names floors *outside* generation — Value of Information (Vol), grammar coverage, Landauer/joule estimate, certificate, settle-refuse — and demotes the neural net to a residual leaf. Soft-ref mol prove (~29 VERIFIED criteria in the mixture-of-limits workspace) is claimable as **constructive existence in software**. This study does **not** claim FPGA or board package energy. Landauer appears only as a labeled estimate. `measured_j` is never invented; `board_synth_claimed=false`.

Claim / non-claim banner

What this study claims

- MoL as a clean-room navigation law (floors + cascade + close/receipt), not a wrapper on MoE.

- Soft-ref constructive existence: in-tree mol prove criteria printing VERIFIED (software reference).
- Information-theoretic / physics-rooted thesis: compression and closed forms beat excess enumeration when grammar is covered.
- Analytical / catalog energy estimates (estimated_j, mu_source=catalog) and Landauer as a **labeled thermodynamic lower-bound estimate**.
- Typed honesty: ModelGenerated cannot coerce to Deterministic; refuse is a lawful success with a receipt.
- **Bottleneck addendum:** awareness of floors already exists in IT/physics/decision theory; the hard gap is applied-math × materials/hardware embodiment (not conceptual ignorance).

What this study does not claim

- Board, FPGA, RAPL, NVML, or package joules (measured_j stays unset in soft-ref; never invent meters).
- board_synth_claimed is **false** — no synthesized or metered MoL board package is reported here.
- That Landauer equals wall joules, OpCounter estimates, or any hardware probe.
- AGI achievement, ARC-AGI leadership, or MoE accuracy SOTA (Leapfrog owns the MoL≠MoE + model[?]model table in §9).
- Live WCA MCP, Ferric/EFA hardware certificates, klere-vm FPGA meters, or the full Periodic Stack 258 live catalog as proven here.

1. Introduction — pursuit of limits toward AGI

The dominant construction for computer intelligence (CI) treats the neural net as the substrate and excess tokens as the growth variable. That construction diverges from the historical method that produced reliable science. When Brahe enumerated planetary positions, the tables were necessary but not the law. Kepler compressed years of positions into three predictive relations. Newton further compressed those relations into invariants and closed forms that travel across domains. The same pattern recurs wherever a grammar of nature is covered: a short formula outruns an infinite ledger of cases.

Information theory names the modern accounting of that pattern. Shannon (1948) prices bits under uncertainty. Value of Information (Vol) asks whether another observation is worth its cost for a decision. Landauer (1961) prices irreversible bit erasure in joules at temperature (T): $(E_{\text{bit}} = k_B T)$ per bit erased in the ideal model — a **lower-bound estimate**, not a wattmeter reading. Together they imply floors: past a point, more bits stop buying outcomes that matter for a stated benefit.

Thesis. Pursuit of limits is the path toward AGI-grade reliability. Excess-token and MoE scaling diverge from true Vol for a given benefit. MoL is the executable navigation law that binds those floors — Vol, grammar, Landauer/joules (estimate), certificate, settle-refuse — against industry Mixture-of-

Experts. **Addendum:** the bottleneck is not conceptual ignorance — Shannon, Landauer, Kolmogorov/Solomonoff/Chaitin, Howard (Vol), and formula-first science AI already spoke to limits — but **applied mathematics embodied in materials and physical hardware**.

This paper's contribution is the study prose for that law as published on research.openie.dev, grounded in the clean-room mixture-of-limits software reference (Apache-2.0 OR MIT). Companion studies already on this site supply the commit-record interface ([Notational Intelligence as Commit Law](#)) and the economic stop after free digital inference ([Satiation and Scarcity after Free AI](#)). MoL is the navigation law those companions sit under: NI owns irreversible commit shape; Satiation owns economic done; MoL owns *which gear closes* and *when refuse is success*.

Three measurement statements constrain every later number. First, joules from catalog surrogates and OpCounter-style analytics are **estimates**, not board power. Second, Landauer annotations are **estimates**, never equal to RAPL/NVML/measured_j. Third, this study sets `board_synth_claimed=false` and does not invent `measured_j`.

Scope. Section 2 states the law and Periodic Stack navigation. Section 3 states the proof spine $(E(x)(D)(S,V))$. Section 4 describes the cascade and close/receipt bind. Section 5 bridges to Satiation without rewriting it. Section 6 is a Leapfrog-owned placeholder for the prove[?]claim map. Section 7 sketches Vol, grammar, and settle-refuse mathematics. Section 8 places related work as a verified citation chain (historical→recent proof points) plus Tier A formula/mechanism systems, and states that the gap is physical/materials + applied-math embodiment — not lack of awareness. Section 9 is a Leapfrog-owned placeholder for $MoL \neq MoE$ and model[?]model SOTA. Section 10 opens toward AGI via limits.

2. Law — floors where more bits stop buying outcomes

Mixture of Limits asserts: for a task coordinate, there exist floors such that additional tokens, parameters, or joules do not purchase verifiable progress. Intelligence is therefore a **navigation law** over a structured stack of primitives — not an invitation to dump residual capacity into a generative corridor.

Consequences:

1. **Mathematical compression / formulas beat excess tokens** when the grammar is covered.
2. **Neural nets are a demoted residual leaf**, not the default substrate of CI.
3. Routing is **cheapest-sufficient**: escalate only on miss; refuse when escalation is unsafe or Vol-negative.
4. Proof / energy law: $(E(x)(D)(S,V))$ with honest joule receipts; Landauer labeled as estimate.
5. **Available devices / multi-fabric**: after a cascade gear closes, pick the cheapest sufficient device class — not one accelerator by default.

Named floors

Limit id	Kind	Meaning
voi	ValueOfInformation	Marginal bits worthless under budget — stop
grammar	GrammarCoverage	Covered by Lookup/Formula/Solver — or refuse unknown
energy	Energy	Joule budget floor; Landauer-aware estimates
efa_certificate	Certificate	Diverge / fail certify → refuse before commit
settle_refuse	SettleRefuse	Ternary / energy-landscape settle will not settle → refuse
primitive_gap	PrimitiveGap	Missing Periodic Stack primitive — surface the gap
safety / wca_refuse	Safety / WCA	Capability deny; WCA certify refuse
information / latency	Information / Latency	Sufficiency and soft wall-clock ceilings

Neural net demotion

The model tier is **LAST**. Default budgets set `allow_model=false`. Even when allowed, the model remains a leaf: it may propose residual content; it never bypasses certify-before-commit, and ModelGenerated answers cannot be laundered as Deterministic. Coin-cell / edge power is $O(1)$ formula plus LUT refuse, not a tiny transformer.

Periodic Stack

MoL navigates the OpenIE **Periodic Stack** (compute.openie.dev: **258 primitives / 33 families**). The soft-ref proves an in-tree **subset** navigator with Present/Gap cells and scale notes citing 258/33. Full live catalog and μ calibration corpora remain out of soft-ref proof scope. Empty cells surface as `primitive_gap` — MoL does not hallucinate coverage.

MoL is not MoE

MoE (generative corridor)	MoL (navigation law)
Experts live <i>inside</i> a model; gate selects parameters	Limits live <i>outside</i> generation; floors stop escalation
More experts → more capacity for the same generative act	More named floors → sharper refuse / cheaper close
Success = next-token likelihood	Success = closed grammar + certified commit + joule receipt

MoE (generative corridor)	MoL (navigation law)
Model is the substrate	Model is a demoted residual leaf

Clean-room policy: MoL is law + runtime, not a wrapper on MoE, and not a path-dependent fork of sibling OpenIE trees for the proven path.

3. Proof spine — (E(x) (D)(S,V))

The energy / impedance law used throughout MoL is:

$$[E(x) (D)(S,V)]$$

where (E(x)) is path energy for closing act (x), ((D)) scales with decision / difficulty structure (D), and ((S,V)) is an impedance factor for stack coordinate (S) and view (V). Soft-ref receipts stamp catalog () with mu_source=catalog and a landauer_floor_ratio. The product is an **analytical / catalog estimate**, not a package joule reading.

Landauer (estimate only). For irreversible erasure of (n) bits at temperature (T),

$$[E_{\{ \}}(n,T) = n,k_B T]$$

Receipts may annotate landauer_floor_J from this formula. That annotation is explicitly **estimate ≠ measured**. Soft-ref closes keep measured_j=None. Optional OS meter features (RAPL / IOReport / powermetrics), when present and successful, may populate measured_j only under a labeled MeasureSource; failure or VM stays unavailable and **never invents** numbers. This study's soft-ref claim set does not depend on those meters.

Historical rhyme. Kepler did not need every future observation once the three laws closed the grammar of planetary motion for the epoch. Newton did not need a larger ephemeris table to predict a new orbit once (F=ma) and inverse-square gravitation covered the coordinate. Shannon did not need infinite samples to bound channel capacity. Landauer did not need a particular chip to state a thermodynamic lower bound. MoL's spine is the same move for CI: bind a floor, close cheapest-sufficient, refuse when the floor says stop.

Constructive existence (software). Soft-ref cargo run -p mol-cli -- prove is required to print VERIFIED per criterion and exit 0 (~29 soft-ref criteria spanning deterministic close, formula/lookup without model, Vol refuse, settle commit+refuse, certificate refuse, capability default-deny, receipt honesty, replay-class coercion deny, Periodic Stack subset navigation, μ catalog, transcript replay, Z2 cite / Z1 compose, agent mailbox, bitemporal memory, fabric routing, desktop headless shell, OS meter honesty, WASM capsule, Agent Lane, multi-fabric receipts, ecosystem e2e certify — see PLAN.md in the MoL workspace). That is claimable as constructive existence **in software**. It is **not** a board energy claim. Detailed prove²claim mapping is Leapfrog-owned (§6).

4. Cascade and close — Lookup → Formula → Solver → Model LAST

MathGround / OpenIE cascade order (type-enforced replay classes):

Lookup → Formula (closed-form) → Solver / settle → Model LAST

MoL tier	Role	Replay class (typical)
Lookup	LUT / allow-table / registry hit	Deterministic
Formula	Closed identity / unit convert / law instance	Deterministic
Solver	Sparse solve / ternary settle	Deterministic or compose under Solver
Model	Stochastic residual only	ModelGenerated

Invariant: ModelGenerated →/Deterministic. Typed answers refuse coercion.

Automation / close loop:

sense → classify grammar → cheapest sufficient → certify → commit|refuse → receipt

MoL **owns close**. Refuse over escalate when unsafe or Vol-negative. Refuse is a **lawful success** and still emits a receipt.

co/0 bind sketch (namespaces)

Soft-ref honesty binds act namespaces in the spirit of OpenIE co/0 receipts (sketch, not a live leapfrog path-dep):

Namespace	Role
claim.*	Seeded / retrieved knowledge claims (Z2); unknown → refuse, no invent
act.*	Capability-gated acts; Mutate/External default-deny
close.*	Route + certify gate; commit or refuse
receipt.*	estimated_j labeled; measured_j only if Metered probe returned; board_synth_claimed=false

Rules stamped on every soft-ref close:

1. **Certify-before-commit** — no irreversible side effect without certificate pass.
2. **ModelGenerated →/Deterministic** — sealed replay markers.
3. **measured_j only if Metered** — otherwise None / unavailable; never invent.

4. **Refuse = lawful success + receipt** — Vol / settle_refuse / certificate / capability denies still close the ledger.

Multi-fabric soft-ref: after tier selection, route cheapest sufficient DeviceKind (Cpu always present; Gpu*/Wasm/ThermoSettle/... optional). Detection ≠ joules.

5. Econ bridge — Satiation (cite, do not rewrite)

Digital inference prices can fall toward free-at-margin without implying free energy, free actuation, or unbounded value after a chore is complete. The companion study [Satiation and Scarcity after Free AI](#) defines satiation as a stop on a written completeness predicate and cites published inference-price series. MoL supplies the *machine* stop that matches that economic intuition: Vol and grammar floors refuse excess generation once benefit is covered; commit/refuse receipts (NI) record the irreversible boundary.

This section does not restate Satiation’s claims or price tables. Read Satiation for economic done; read NI for the commit record; read MoL for navigation floors that make “done” and “unsafe” executable without opening the model leaf.

6. Prove claim map

MoL’s constructive prove is not a leaderboard stunt. It is an **information-theoretic** close law rooted in **physics**: the same historical arc that took astronomy from excess epicycles to Kepler/Newton predictive floors, and that takes computation from Shannon → Landauer → complexity / Vol toward named energy floors. Global academia already holds those proof points; the bottleneck is **applied math × materials (hardware) embodiment**, not awareness of the slogans. This section only maps what moL prove actually stamps VERIFIED onto paper claim.* ids — embodiment of known floors, not invention of new physics.

A claim is **proven** only when the in-tree harness prints VERIFIED for the named criterion and exits 0. A claim is **soft-ref only** when PLAN §6 marks it OUT OF PROOF SCOPE, or when the criterion itself stamps soft-ref / fixture / feature-gated behavior rather than live silicon. No row invents measured_j, board joules, or board_synth_claimed=true.

Measurement labels used below (OpenIE honesty grammar):

Label	Meaning
Unmetered	Software-ref / soft-ref path; measured_j=None; estimate may be labeled
Estimated	Catalog / Landauer / fuel / μ surrogate with explicit EstimateKind or mu_source=catalog

Label	Meaning
Metered	Optional live probe (energy-meter) actually returned numbers — not required for prove; never invented

Default close path for all proven claims: **Unmetered** (+ **Estimated** where μ / Landauer / fuel appear).
board_synth_claimed=false everywhere in the proven path.

6.1 Harness meta

Prove #	Criterion (PLAN)	Paper claim id	Status	Measurement
P0	cargo test --workspace green	claim.mol.ci_workspace	Proven (CI / operator; not printed by mol prove)	—
P1	cargo run -p mol-cli --prove exits 0	claim.mol.prove_harness	Proven (prints VERIFIED per row below)	Unmetered

6.2 Proven claims (P2–P24)

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
P2	Deterministic close (double-run)	claim.mol.deterministic_close	Same asks yield the same commit/refuse + limit id.	—	Un
P3	Formula/lookup close without model	claim.mol.formula_lookup_no_model	Landauer / unit-convert closes COMMIT at Lookup or Formula; model never answered.	—	Un Est (La floc rec
P4	Vol refuse when !allow_model	claim.mol.voi_refuse	Free-form ask → REFUSE voi.	—	Un
P5	Settle commit + settle refuse	claim.mol.settle_ternary	Ternary settle → COMMIT; will not settle → REFUSE settle_refuse.	Software-ref settle stub; no klere-vm / FPGA	Un
P6	Certificate refuse on diverge	claim.mol.efa_certificate_refuse	Formula + diverge → REFUSE efa_certificate.	In-tree EFA-style cert; Ferric / live BMI	Un

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
				loop soft-ref	
P7	Capability default-deny mutate	claim.mol.capability_deny_mutate	AutomateGate::default().gate(Mutate) → Refuse.	—	Un
P8	Receipt honesty	claim.mol.receipt_honesty	Software-ref: measured_j=None, board_synth_claimed=false, estimate labeled (EstimateKind).	Live RAPL/ NVML default close OUT OF SCOPE	Un Est
P9	ModelGenerated →/ Deterministic	claim.mol.replay_no_strengthen	TypedAnswer::weaken_to returns ReplayCoercion.	—	—
P10	Periodic Stack subset navigation	claim.mol.stack_subset_nav	Family + present primitive + scale → COMMIT at Lookup; note cites 258/33.	Full 258 live catalog OUT OF SCOPE	Un
P11	primitive_gap via registry probe	claim.mol.primitive_gap	Gap marker physical_settle + absent name → REFUSE primitive_gap (not string-only).	—	Un
P12	μ / impedance catalog	claim.mol.mu_catalog	Receipts stamp mu_source=catalog, mu, landauer_floor_ratio; $E \approx \theta \cdot \mu$ (not fake RAPL).	μ calib corpus OUT OF SCOPE	Est
P13	Receipt transcript replay	claim.mol.receipt_replay	JSONL/in-memory replay reproduces commit/refuse + limit id; model never answered.	—	Un
P14	Z2 retrieve+cite	claim.mol.z2_retrieve_cite	Seeded factual hit → COMMIT RetrievedCited + citation ids; unknown → REFUSE claim_unknown (no invent).	Seed corpus in-tree only	Un
P15	Z1 compose/ synthesis	claim.mol.z1_compose	≥ 2 cited claims → COMMIT Composed + synthesis.composed_from; missing → REFUSE compose_missing; never laundered as Deterministic/RetrievedCited alone.	—	Un

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
P16	Agent mailbox loop	claim.mol.agent_mailbox	Goal/Message/Act → close → commitl refuse + receipt in transcript; demo hits formula/cite/compose COMMIT and voi/ settle_refuse REFUSE; model cold; honesty flags hold.	—	Unr
P17	Bitemporal state + memory	claim.mol.bitemporal_memory	Writes only on MoL close COMMIT; Mutate deny / free-form remember → REFUSE; recall → RetrievedCited + memory: cite; unknown → memory_unknown.	—	Unr
P18	Fabric routing + soft-ref inventory	claim.mol.fabric_soft_ref	Offline inventory valid (source=software_ref, Cpu always); Lookup/Formula→Cpu; Settle→ThermoSettlelCpu; Model→Gpu* or REFUSE fabric_unavailable; detect ≠ measured joules.	Live wgpu presence optional / not required for prove	Unr
P19	Desktop shell headless	claim.mol.desktop_headless	ShellSession ask/close → receipt view (zone/fabric/limit/estimated_j/ measured_j=None); joule ledger; Vol refuse; GUI not required.	egui GUI optional	Unr Est
P20	OS meter honesty	claim.mol.os_meter_honesty	Feature off / VM / no permission → measured_j=None, empty components, measure_source=unavailable; fixtures stamp labeled sources without inventing; default close stays unmetered.	Live probe optional (energy-meter); never invent	Unr (de Me whe retu
P21	WASM capsule certify	claim.mol.wasm_capsule	add.wasm under stub runtime; fuel → estimated_j only; FS/net grant / fuel exceed / host-share → REFUSE; close stamps capsule.	Optional wasmtime feature	Unr Est (fue
P22	Agent Lane session path	claim.mol.agent_lane	Lane partitions; host_invoke without provenance refuses; keyword confirm insufficient — need GrantReceipt; allow stamps AgentLaneReceipt.	—	Unr
P23	Multi-fabric compute receipt	claim.mol.fabric_compute_receipt	Soft-ref inventory; ComputeFabric / ScheduleDecision / ComputeStepReceipt with fabric_id; cascade stamps fabric:route; CPU Formula COMMIT; Model residual may fabric_unavailable; never invent measured_j.	Ferric not path-dep'd	Unr

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
P23b	Live fabric soft	claim.mol.fabric_live_soft	Feature off stays software_ref; mock/live Metal can stamp gpu_metal; Formula stays Cpu; missing Gpu* fail-closed; prove does not require a GPU; measured_j never invented.	Soft / optional detect	Un
P23c	wgpu / Metal tiny kernel	claim.mol.kernel_vector_add	Clean-room WGSL vector-add; soft stub offline (same checksum); Gpu* stamps kernel:vector_add + execution_proof; optional overlapping meter may stamp package measured_j with energy_honesty=measured — never invent / never rail-sum.	Live Metal optional; fixture OK for prove	Un (de Me if m retu
P24	Ecosystem e2e certify (one receipt)	claim.mol.ecosystem_e2e	Lane → fabric (Cpu soft-ref) → WASM → energy honesty (estimated fuel; optional meter_sample) → host invoke only with GrantReceipt → commitIrefuse on one receipt; refuse-without-grant → grant_receipt_required.	—	Un Est

6.3 Soft-ref only (not proven; do not escalate to paper results)

These are PLAN §6 residuals. Citing them is allowed as roadmap / reference semantics. Treating them as measured results is not.

Soft-ref item	Why soft-ref	Related prove row (if any)
Live silicon RAPL / NVML as default close	Optional energy-meter; soft-ref prove stays measured_j=None	P8, P20
Live wgpu adapter presence	Feature fabric-detect; inventory soft-ref remains prove default	P18, P23b, P23c
Ferric / on-device EFA (BMI hardware loop)	Reference semantics only; not path-dep'd	P6, P18, P23*
klere-vm WASM / FPGA meter (real pJ)	Software-ref StubKlereSettle	P5
Live openie-path / leapfrog ask bridge	Adapter port	—
Live WCA MCP / wca-lut-edge in-proc certify	Adapter port	—
Full Periodic Stack 258 live catalog + μ calib corpus	Subset navigator + Gap + tier μ catalog are in proof	P10, P12

Soft-ref item	Why soft-ref	Related prove row (if any)
Trained weights / candle / tract Model leaf	Model stays demoted stub	P3, P4, P16

6.4 Seeded knowledge claims (Z2 corpus; not the prove map)

In-tree ClaimStore seeds used by P14/P15 (retrieve/compose), distinct from the paper claim ids above:

- `claim:landauer.principle`
- `claim:mol.law`
- `claim:mol.proof_law`
- `claim:mol.cascade`
- `claim:stack.scale`
- `claim:honesty.no_fake_rapl`
- `claim:replay.retrieved_cited`
- `claim:openie.z2`

6.5 What this map refuses

- No invented `measured_j` values.
- No `board_synth_claimed=true`.
- No board joules / FPGA watt leadership from MoL v0.1 prove.
- No promotion of soft-ref roadmap items into proven claims.

7. Math sketches — Vol stop, grammar coverage, settle-refuse

7.1 Vol stop

Let (B) be a benefit functional for a decision, (x) the current information state, and (c) the cost (bits, latency, or estimated joules) of acquiring observation (y). A simple stop rule:

$$[(x,y) = [B \ x,y] - [B \ x] - , c(y)]$$

Refuse escalation when $((x,y))$ (or $()$ for a configured threshold). Soft-ref MoL encodes this as floor `voi` with `allow_model=false` by default: free-form asks that would only open the model leaf refuse rather than spend. This is the Shannon/decision-theoretic cousin of Kepler stopping new naked-eye points once the law predicts within tolerance — not a claim that soft-ref computes a calibrated Bayesian Vol for every domain.

7.2 Grammar coverage

Let (G) be the covered grammar (Lookup registry U Formula identities U Solver settle programs). A query (q) is **covered** if a deterministic derivation (q _G a) exists with replay class not ModelGenerated. Cascade order tries Lookup, then Formula, then Solver. If covered, **do not open the model**. If uncovered and policy forbids model escalation, refuse (grammar / voi / primitive_gap as applicable). Historical analog: once Newton’s grammar covers orbital mechanics for the force law at hand, one does not re-enumerate Brahe’s table to answer a new initial-condition query.

7.3 Settle-refuse

Under Solver, a ternary / energy-landscape settle either reaches a certified fixed point (commit) or signals will-not-settle (refuse settle_refuse). Soft-ref implements this as real in-tree settle logic plus certificate diverge tags (efa_certificate). Inspiration from thermodynamic settle is **conceptual**; MoL v0.1 does not claim annealer drivers or adiabatic wall joules. Landauer still appears only as labeled estimate on receipts.

7.4 Bound sketch tying Shannon and Landauer

For (n) irreversible erasures on a path that closes a decision,

$$[E_{\{ \} } E_{\{ \} } (n, T) = n, k_B T]$$

and MoL’s catalog estimate further requires (E_{\{ \} }(x) (D)(S, V)). Neither inequality licenses reporting measured_j without a meter. Excess tokens that do not change (B) fail the Vol test even when joule budgets remain — the information floor binds first.

8. Related work — proof points, lineage, and Tier A

8.0 Academic proof points: top minds already named the floors

The MoL thesis is not a discovery that “limits exist.” It is an executable navigation law for a fact that information theorists, physicists, and decision theorists already stated — often decades ago — and that formula-first science AI keeps rediscovering. Below is a **historical** → **recent citation chain** with DOIs verified for this draft (or marked TBD). These are global academic proof points that compression, floors, and formulas beat excess enumeration.

Information, complexity, and physical cost

Proof point	Who / when	What they already said	Cite (verified)
Bits under uncertainty	Shannon (1948)	Quantitative communication limits; entropy as missing information	10.1002/j.1538-7305.1948.tb01338.x (Part I); Part II 10.1002/j.1538-7305.1948.tb00917.x

Proof point	Who / when	What they already said	Cite (verified)
Irreversible erasure costs heat	Landauer (1961)	Logical irreversibility $\Rightarrow$ minimal heat $\sim(kT)$ per irreversible function	10.1147/rd.53.0183
Reversible computing	Bennett (1973)	Computation can be logically reversible; Landauer cost is about erasure, not “computing” per se	10.1147/rd.176.0525
Algorithmic complexity	Kolmogorov (1965/1968)	Quantity of information as shortest program length — compression as definition	Russian original <i>Probl. Peredachi Inf.</i> 1(1):3–11 (1965), no DOI; English transl. 10.1080/00207166808803030 ; IEEE note 10.1109/TIT.1968.1054210
Algorithmic probability / induction	Solomonoff (1964)	Prefer short programs; universal prior over computable regularities	Part I 10.1016/S0019-9958(64)90223-2
Algorithmic incompleteness	Chaitin (1977)	Algorithmic information bounds what finitely axiomatized theories can prove	10.1147/rd.214.0350
Probability as extended logic	Jaynes (2003 book; earlier papers)	Maximum-entropy / logic-of-science discipline against ad-hoc excess modeling	10.1017/CBO9780511790423
Textbook IT + decision links	Cover & Thomas	Canonical Elements of Information Theory (2nd ed.)	Wiley book DOI 10.1002/047174882X
Value of Information	Howard (1966)	Shannon bits $\neq$ decision value; Vol joins probability with economic consequence — stop when information is worthless for the decision	10.1109/TSSC.1966.300074
Physics of computation lectures	Feynman (pub. Hey/Allen eds.)	Physical limits of computing, reversible/thermo themes in lecture form	Anniversary ed. 10.1201/9781003358817 ; 1996 Addison-Wesley ISBN 0-201-48991-0

Historical mathematical physics (compression beats enumeration) — see §8.1:

Brahe→Kepler→Newton is the pre-IT proof that tables yield to laws. Soft-ref MoL does not invent new archival DOIs for *Principia*; imprint DOIs for critical editions remain **TBD** if a journal version requires them.

Recent formula / mechanism / certify systems (Tier A inventory) — from docs/adjacent-field-hunt.md and docs/appendix-mwm-ai-newton.md (2026-09-30):

Proof point	Who	MoL reading	Cite (verified)
AI Feynman	Udrescu & Tegmark	NN decomposes; closed form is the commit	10.1126/sciadv.aay2631
SINDy	Brunton / Proctor / Kutz	Sparse grammar; reject dense residual fits	10.1073/pnas.1517384113
PySR	Cranmer	Production symbolic regression	arXiv: 2305.01582 (journal DOI TBD if distinct)
AI-Newton	Fang et al.	Concept library + general laws across experiments	arXiv: 2504.01538
AlphaGeometry / AlphaProof	DeepMind	Propose → symbolic/Lean certify (hard floor)	Nature 10.1038/s41586-025-09833-y
Mechanistic World Models	Posner / Lei / Schölkopf	Mechanisms > predictive MoE modules	arXiv: 2607.12474
AutoSINDy hybrid	(hunt 2026 note)	PySR → library → SINDy	arXiv: 2605.09696

Raiffa-style decision analysis texts are the pedagogical cousins of Howard's Vol; a specific Raiffa imprint DOI is **TBD** here rather than invented.

8.0.1 The gap is embodiment — not awareness

Claim (addendum). Applied mathematics × materials science (physical hardware) is the bottleneck, **not** awareness of what should be done.

Evidence for the awareness side is the table above: Shannon priced bits; Howard priced Vol for decisions; Landauer/Bennett priced irreversible erasure; Kolmogorov/Solomonoff/Chaitin priced shortest programs and incompleteness; Feynman taught physical limits of computation; SINDy/PySR/AI Feynman/AI-Newton/AlphaGeometry/MWMs keep showing that formulas, sparsity, verification, and mechanisms beat unbounded enumeration and unguided MoE-style prediction.

What remains hard — and what MoL's companions (NI commit law; coin-cell Formula+LUT; multi-fabric routing; Metered vs estimated honesty) point at — is **embodying** those floors in real stacks: materials and devices that make Lookup/Formula/Solver cheap, certificates enforceable before irreversible acts, and joule receipts honest without inventing measured_j. Soft-ref moL prove shows the *law+runtime* can exist in software. It does not claim the materials problem is solved. board_synth_claimed=false; no fabricated board package energy.

8.1 Historical lineage — compression beats enumeration

Tycho Brahe → Kepler. Brahe's program was high-fidelity enumeration. Kepler's three laws compressed those tables into predictive relations. Enumeration was necessary input; the law was the compressible object that stopped needing every future observation.

Kepler → Newton. Newton folded planetary regularities into invariants and closed forms ($F=ma$); universal gravitation). Predictive power migrated from case tables to formulas that travel. AI-Newton (Fang et al., arXiv:2504.01538) is a contemporary soft echo: concept libraries and general laws across many noisy mechanics experiments — NN as recommender residual, not substrate.

Invariants and closed forms. Symmetry → conservation (Noether tradition), Hamiltonian/Lagrangian reduction, and integrals of motion are the mathematical-physics statement of “formula wins when grammar is covered.” MoL's Formula gear is that tradition's CI leaf.

Information theory and physical cost. Shannon, Howard (Vol), Landauer, Bennett, Kolmogorov/Solomonoff/Chaitin, and Cover & Thomas form the IT/decision/thermodynamic chain in §8.0. MoL binds them as named floors and labeled estimates — not as a claim of operating near the Landauer bound in silicon.

Notational intelligence. Lee's notational-intelligence claim is the companion frame for executable predicates; see [NI Commit Law](#).

8.2 Tier A detail — formula / mechanism / executable physics

Inventory dated 2026-09-30 in the MoL workspace hunt docs. Capsule readings:

1. **AI Feynman (+ 2.0)** — Physics-inspired symbolic regression; NN for symmetries/separability, then recursive formula recovery. Canonical *Formula* gear.
2. **SINDy** — Sparse identification of nonlinear dynamics; explicit grammar + sparsity floor.
3. **PySR** — Evolutionary symbolic regression; distill NNs → closed forms.
4. **AI-Newton** — Concept-driven law discovery across noisy mechanics experiments.
5. **AlphaGeometry / AlphaProof** — LM proposes; verifier is the hard floor (propose → certify → commit/refuse template).
6. **Mechanistic World Models** — Sister philosophy: reusable mechanisms; MoE as wrong module semantics for explanation.

Additional hunt rows (AtomAgents/SciAgents, KeplerAgent/NewtonBench, AI-Descartes, DreamCoder/COMET/NEO, Robot Scientist) sit as hybrid or mechanism-library cousins. Where a DOI was not verified beyond arXiv IDs, this study cites arXiv and marks journal DOI **TBD**.

MoL reading of Tier A. Formula/mechanism discovery (A) plus physics-executable leaves (B in the hunt) beat token-escalation workflows (C). MoE-style mixtures are the wrong “Mixture of.”

8.3 What MoL adds

Tier A systems recover or verify formulas. Classical IT already priced bits, Vol, and erasure. MoL names the **navigation law** that decides *when* Lookup/Formula/Solver suffice, *when* to refuse, and *how* receipts stay honest about estimated vs measured joules — then implements a clean-room soft-ref close loop. Soft-ref constructive existence is software, not a materials breakthrough and not a claim that MoL rediscovered Newton II on a new corpus.

9. MoL ≠ MoE, and the model?model SOTA table

9.0 Spine (history + mathematics, not a horse race)

MoL is proven as an **information-theoretic** approach rooted in **physics**: formula/law discovery in the Newton–Kepler lineage, not excess generation. Academia already has Shannon → Landauer → complexity/Vol → symbolic and energy floors; MoL **embodies** those floors in a commit/refuse cascade. The remaining bottleneck is applied math × materials (hardware), not an awareness gap. The table below situates field systems so readers see what they optimize — **which neural generator runs** — without turning MoL into another SOTA horse race. MoL asks whether a generator should run at all, converging to Vol/floors the way physics converged to predictive laws.

9.1 Law contrast (not a rebrand)

Mixture of Experts (MoE) places experts **inside** a generative corridor: a gate selects parameters; success is next-token likelihood; capacity grows by adding experts to the same act. Mixture of Limits (MoL) places named **floors outside** generation: Lookup → Formula → Solver/settle → **Model LAST**; success is closed grammar + certified commit/refuse + joule receipt; the model is a demoted residual leaf, cold by default (`allow_model=false`).

Axis	MoE / model?model field	MoL
Where the mixture lives	Inside the model (experts, draft/target pairs, strong/weak LLMs)	Outside generation (named floors + cascade gears)
Routing object	Which model (or draft head) answers	Which tier closes: Lookup / Formula / Solver / Model
Refuse	Rare; usually escalate to a larger model	First-class: Vol, grammar, settle, certificate, primitive_gap, fabric_unavailable, ...
Energy honesty	Often latency/cost proxies; some papers meter GPU energy for LLM serving	Receipts: estimated_j labeled; soft-ref measured_j=None; board_synth_claimed=false; never invent RAPL
Success metric	Tokens/s, cost@quality, speculative speedup	Certified commit + typed ReplayClass + receipt

MoL does not claim to beat RouteLLM on MT-Bench cost, GreenServ on Wh, or HCSpec on decode speedup. Those systems optimize **which neural generator runs**. MoL asks whether a generator should run at all.

9.2 Field SOTA (all model[?]model routing / speculation / constrained generation)

Every row below is **model[?]model**: routers choose among LLMs; speculative methods draft-then-verify with draft/target models; constrained decoding still samples from an LLM under a grammar. None implements Lookup→Formula→Solver→Model LAST with refuse-to-commit as the close law.

Measurement labels: **Metered** | **Estimated** | **Unmetered** — only when that paper’s own method section supports the label. MoL does **not** re-report their joules as MoL measured_j. MoL prove path: **Unmetered** (software-ref); board_synth_claimed=false; no invented board joules.

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs MoL
RouteLLM	<i>RouteLLM: Learning to Route LLMs with Preference Data</i> (arXiv); ICLR 2025 venue title uses <i>from Preference Data</i>	arXiv: 2406.18665 · ICLR 2025	Preference router (strong [?] weak LLM)	Query → strong or weak LLM	Cost / quality (API \$); not MoL Landauer receipts	Model [?] model; no Lookup/Formula floor; no typed refuse-to-commit
PEARL (routing)	<i>PEARL: Performance and energy aware routing for LLMs</i>	DOI: 10.1016/j.future.2025.108218 · FGCS 176:108218	Energy-aware multi-LLM router	Query → LLM under energy cap (EMM predicts energy)	Authors report energy-aware routing (GPU/infra); not imported as MoL measured_j	Still picks a model; MoL may refuse before any model
GreenServ	<i>GreenServ: Energy-Efficient Context-Aware Dynamic Routing for</i>	arXiv: 2601.17551 · ICPE 2026	Contextual bandit multi-LLM router	Query features → LLM; accuracy vs measured GPU energy (Zeus)	Metered (authors’ GPU Wh via Zeus) on their LLM pool — field result, not	Model pool routing; MoL cascade is non-neural first

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs MoL
	<i>Multi-Model LLM Inference</i>				MoL board synth	
VoltanaLLM	<i>VoltanaLLM: Energy-Efficient and SLO-Aware Disaggregated LLM Serving via Adaptive Frequency Control and State-Space Routing</i>	arXiv: 2509.04827	P/D-disagg serving + frequency + state-space route	Prefill/decode instances + GPU frequency under TTFT/ITL SLOs	Metered (authors; pyNVML on A100/GH200) up to ~36.3% E2E GPU energy vs max-freq baseline — field serving result	Routes instances of the same generative stack ; MoL may never open decode
HCSpec	<i>HCSpec: Two-Tier Horizontal Cascade Speculative Decoding for High-Efficiency Large Language Model Inference</i>	DOI: 10.18653/v1/2026.acl-long.353 · ACL 2026	Speculative decoding (draft [?] target)	Position-specialized draft cascade → target verify	Latency speedup (vs EAGLE-3 / AR); energy not MoL's claim	Speeds generation; does not refuse generation on floors
CAS-Spec	<i>CAS-Spec: Cascade Adaptive Self-Speculative Decoding for On-the-Fly Lossless Inference Acceleration of LLMs</i>	arXiv: 2510.26843 · NeurIPS 2025	Self-speculative cascade (DSIA drafts)	Draft stages from target (sparsity/quant) + DyTC	Latency speedup ~1.1×–2.3× AR (authors); lossless tokens	Model-internal draft hierarchy ≠ MoL floors
GCD	<i>Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning</i>	arXiv: 2305.13971 · DOI: 10.18653/v1/2023.emnlp-main.674 · EMNLP 2023	Grammar-constrained LLM decoding	Mask logits to CFG	Unmetered / quality metrics	Constrains tokens ; MoL constrains commit

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs MoL
CRANE	<i>CRANE: Reasoning with constrained LLM generation</i>	arXiv: 2502.09061 · ICML 2025 (PMLR v267)	Reasoning-augmented constrained decoding	Alternate unconstrained reason [?] constrained answer	Accuracy on GSM-symbolic / FOLIO; not MoL joules	Still LLM generation under grammar; MoL can close at Formula without an LLM

Name collision (do not conflate): ICLR 2025 also has *PEARL: Parallel Speculative Decoding with Adaptive Draft Length* (arXiv:[2408.11850](#)) — speculative, not the FGCS energy router. §9’s PEARL row is the **routing** paper (DOI 10.1016/j.future.2025.108218). Speculative PEARL sits in the same [model\[?\]model](#) bucket as HCSpec/CAS-Spec if cited later.

9.3 What MoL claims instead

From PLAN prove (see §6):

1. **Cascade order is law**, not a heuristic over LLMs: Lookup → Formula → Solver/settle → Model LAST.
2. **Refuse-to-commit** is typed and receipted (voi, settle_refuse, efa_certificate, primitive_gap, fabric_unavailable, claim_unknown, ...).
3. **Receipt honesty** on the proven path: measured_j=None (Unmetered), estimates labeled (**Estimated**), board_synth_claimed=false. No invented measured joules; no board joules.
4. **ReplayClass** cannot strengthen: ModelGenerated →Deterministic.
5. Soft-ref multi-fabric routing chooses DeviceKind after tier close — still not “pick GPT-4 vs Mixtral.”

9.4 Honesty fence

Forbidden in MoL paper results	Allowed
Invented measured_j	Soft-ref prove with measured_j=None
board_synth_claimed=true without meter + synth log	board_synth_claimed=false
Importing GreenServ / VoltanaLLM Wh as MoL board energy	Citing them as field model[?]model SOTA with their own Metered labels
Claiming MoL wins tokens/J vs HCSpec	Claiming MoL can refuse before tokens

Field SOTA above remains **model[?]model**. MoL’s contribution is not a better router among generators: it is the navigation law that embodies known information/energy floors, with a prove harness that keeps Model last and refuse first-class.

10. Toward AGI via limits — open questions and companions

If AGI-grade reliability means closing tasks with certificates, receipts, and known floors — not maximizing tokens under an unlimited generative prior — then the historical method already points the way. Kepler and Newton pursued limits: the shortest predictive law that covered the grammar. Shannon priced bits; Howard priced Vol; Landauer and Bennett priced irreversible erasure; Kolmogorov, Solomonoff, and Chaitin priced compression and incompleteness. Recent formula/mechanism systems (AI Feynman, SINDy, PySR, AI-Newton, AlphaGeometry, MWMs) keep proving the same pattern in modern form.

The missing piece is not the idea. Top minds in information theory and precursors to computer automation already spoke to floors. The bottleneck MoL asserts is **applied mathematics embodied in materials and physical hardware**: devices, fabrics, and meters that make cheapest-sufficient close real at the edge — without laundering estimates as board watts, and without treating MoE capacity as a substitute for grammar coverage.

MoL says CI should navigate those floors explicitly: Lookup and Formula first, Solver when settle is the right physics, Model last, refuse when Vol or safety says stop. Soft-ref prove shows the navigation law can be constructive in software. Companions on this site (NI commit record; Satiation economic done) supply interfaces and economic stops. None of that flips board_synth_claimed or invents measured_j.

Open questions

- Full Periodic Stack 258 live catalog and calibrated () corpora (subset navigator is in soft-ref proof; full catalog is not).
- Live WCA MCP / System One pre-gate and leapfrog openie-path ask (adapters remain stubs on the proven path).
- Materials / device paths that make Formula+LUT coin-cell closes dominate generative spend — without fake meters.
- When — if ever — a Metered measured_j path is published beside soft-ref estimates without laundering estimate as board watts.
- How Vol thresholds should be set per domain without smuggling engagement metrics as completeness (see Satiation).
- Whether mechanism libraries (MWM cousins; AI-Newton concept base) should be imported as typed Lookup entries without breaking clean-room prove.

Companions on this site

- [Notational Intelligence as Commit Law](#) — commit/refuse record; analytical energy; board honesty.
- [Satiation and Scarcity after Free AI](#) — economic done after free digital inference.
- Living figures: [/living/mol/](#) (stubs; analytical OpCounter ≠ board power; no fake meters).

Closing sentence. Pursuit of limits — compression into predictive law, priced information, thermodynamic honesty — is the through-line from Kepler and Newton to Shannon, Howard, Landauer, and Kolmogorov. Mixture of Limits is that through-line stated as a navigation law for computer intelligence; embodying it in materials remains the hard problem.

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Kepler's laws / Newton's *Principia* / Brahe's observational program are treated as historical common knowledge in §8.1; imprint DOIs for critical editions **TBD** if a journal version requires them. Raiffa decision-analysis imprint DOI **TBD**.

Appendix A. Honesty checklist

- board_synth_claimed=false
- Soft-ref: measured_j=None; never invent meters
- Landauer = labeled estimate only
- Analytical / catalog estimated_j ≠ board power
- Soft-ref mol prove ~29 VERIFIED = constructive existence in **software** only
- Clean-room MoL: law+runtime, not MoE wrapper
- §§6 and 9 Leapfrog placeholders intentionally incomplete
- DOIs only when verified via search/source inventory; otherwise marked TBD
- Bottleneck claim is materials/applied-math embodiment — not “nobody knew limits exist”

Appendix B. Reproducibility pointers

```
# On a machine with the mixture-of-limits workspace:
cargo test --workspace
cargo run -p mol-cli -- prove # expect VERIFIED lines, exit 0
```

Site preview for this study (research-openie-web):

```
cd /Users/dcharlot/data-share/vibe-coding/research-openie-web
pnpm install
pnpm dev
# open /papers/mol/ , /about/mol , /living/mol/
```

PDF regeneration (optional; needs pandoc + weasyprint): `./scripts/generate-pdfs.sh` after mol is added to the script's `gen_one` list.